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Review Paper

Atomic Layer Etching of SiO_2 for Nanoscale Semiconductor Devices: A Review

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ABSTRACT

In this paper, atomic layer etching (ALE) processes for SiO_2 are reviewed and categorized into two distinct group of anisotropic and isotropic ALE processes. Anisotropic ALE typically involves the fluorination of the silicon dioxide (SiO_2) surface by fluorocarbon deposition during surface modification, followed by the removal of fluorinated layers by relatively low-energy ions. The impacts of the precursor, ion energy, selectivity, and chamber wall conditions on anisotropic ALE processes are reviewed. Isotropic ALE involves the conversion of SiO_2 surfaces into a fluorinated layer or ammonium salt. This layer is subsequently eliminated through various chemical reactions, such as sublimation, fluorination, and ligand exchange. The mechanisms of etching in isotropic ALE are reviewed and classified into two subcategories of thermally isotropic and plasma-assisted isotropic ALE.

Keywords: Anisotropic etching, Atomic layer etching, Isotropic etching, Silicon oxide

1. Introduction

Silicon dioxide (SiO_2) has been applied as an insulator in various nanoscale semiconductor devices, playing a crucial role in the semiconductor industry for the past 50 years [1,2]. SiO_2 exhibits exceptional insulating performance, a bulk resistivity of 10^{15} Ω cm, and a dielectric breakdown strength of 10^7 V cm^{-1}. Additionally, it is cost effective and easy to manufacture and demonstrates excellent compatibility with silicon bulk [3].

Plasma etching is a crucial and essential processing technique in semiconductor device fabrication. Its application to next-generation semiconductor devices is becoming increasingly challenging as the critical dimension (CD) of semiconductors decreases to 10 nm level [4–7]. With decreasing CD and the adoption of three-dimensional structures, conventional reactive-ion etching processes are facing limitations in thickness controllability, etch selectivity, and surface roughness at the nanoscale [6–8]. Consequently, atomic layer etching (ALE) processes are under active development, offering atomic-level precision in layer removal, minimized surface roughness, and exceptional uniformity [4–15].

ALE is a cyclic process that facilitates the atomic-level removal of various layers through a modification step involving radicals or molecules, followed by a removal step utilizing ions or chemical reactions as shown in Fig. 1. ALE processes can provide precise thickness control, excellent surface roughness, and high uniformity at both atomic and nanometer scales [16–21]. A typical ALE process comprises four steps. Initially, the precursor chemisorbs onto the substrate surface through a chemical reaction, which may or may not be self-limiting. The second step involves purging to remove any physically adsorbed reactants. In the third step, modified surface layers are removed, forming volatile etch products via energetic ions or chemical reactions. It is classified as anisotropic ALE when the products are removed faster in one direction by directional energetic ions and as isotropic ALE when removal occurs uniformly in all directions. The final step involves purging the chamber with inert gases, similar to the second step. This sequence is then repeated in subsequent cycles.

In this review, recent advancements in the ALE of SiO_2 are categorized into anisotropic and isotropic ALE processes [8,22–43]. For anisotropic SiO_2 ALE, we summarize the effects of precursors, ion energy, selectivity, and chamber wall conditions. Isotropic SiO_2 ALE processes are reviewed in terms of various etching mechanisms such as

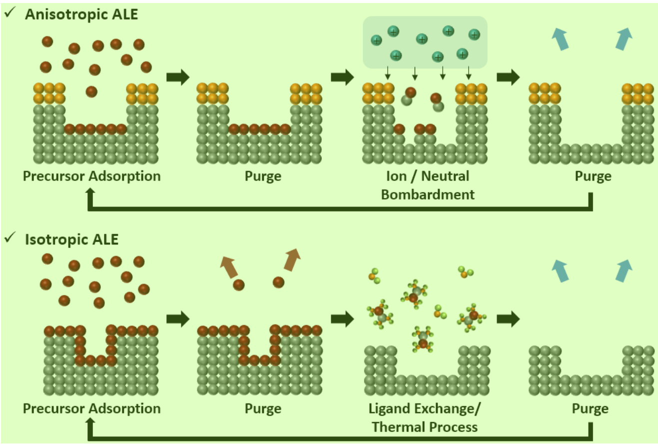

Figure 1. Comparative schematic of anisotropic and isotropic ALE processes for SiO_2.

Table I. Summary of fluorocarbon precursors utilized in anisotropic ALE of SiO_2.

Precursor chemistries for fluorination	Removal	Process temp. (°C)	Etching rate (Å/cycle)	Ref.
C_4F_8/Ar plasma CHF_3/Ar plasma	Ar plasma Ar plasma	10	2.5 3.5	[24]
C_4F_6/Ar plasma	Ar plasma	−10	13	[40]
C_4F_8/Ar plasma C_4F_8/H_2/Ar plasma $C_3H_3F_3$/Ar plasma	Ar plasma Ar plasma Ar plasma	10	2.6 1.6 1.2	[31]
CHF_3 plasma	O_2 plasma Ar plasma	RT	6.8 4.0	[30]
CF_3I/Ar plasma	O_2 plasma	40	9.3	[41]
C_4F_8 plasma CHF_3 plasma n-$C_3F_7OCH_3$ plasma	Ar plasma O_2 plasma	RT	5.8 4.1 2.1	[8]
n-$C_3F_7OCH_3$ plasma i-$C_3F_7OCH_3$ plasma $CF_3CF_2CF_2CH_2OH$ plasma	Ar plasma	RT	2.1 1.8 5.2	[43]

surface modification and the formation of volatile products, encompassing conversion, fluorination, and ligand exchange.

2. Anisotropic SiO_2 ALE

2.1. Fluorocarbon precursors

For anisotropic SiO_2 ALE, the SiO_2 surface undergoes fluorination through the deposition of a fluorocarbon layer, using a range of fluorocarbon-based precursors, as summarized in Table I. Various precursors, including C_4F_8, C_4F_6, CHF_3, $C_3H_3F_3$, CF_3I, and $C_4H_3F_7O$ isomers, are employed in SiO_2 ALE. The chemical composition of the deposited fluorocarbon film, influenced by the precursor type, significantly impacts the etching characteristics. Studies have compared the ALE processes using C_4F_8/Ar and CHF_3/Ar plasmas, focusing on fluorocarbon and hydrofluorocarbon precursors [24]. These studies analyze the composition of the fluorocarbon film and the etching rate in relation to the C_4F_8 and CHF_3 precursors, establishing a correlation between the F/C ratio in the fluorocarbon film and the SiO_2 etching rate with different precursors.

The addition of hydrogen atoms is known to reduce the fluorine radicals in fluorocarbon plasmas, whereas oxygen atoms enhance it [44–46]. SiO_2 ALE processes using C_4F_8, C_4F_8/H_2, and $C_3H_3F_3$ plasmas were investigated to determine the impact of hydrogen addition [31]. The fluorocarbon films generated by C_4F_8/H_2 and $C_3H_3F_3$ plasmas exhibited a lower F/C ratio than those produced by C_4F_8 plasma. Consequently, the SiO_2 etching rate was higher in the C_4F_8 plasma

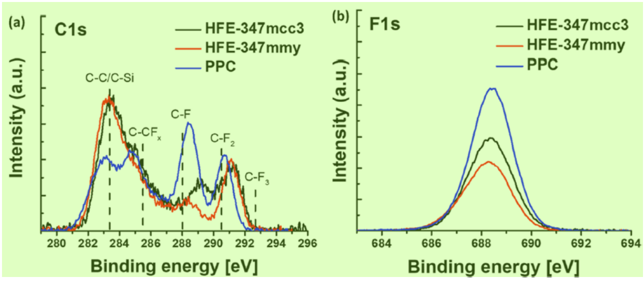

Figure 2. XPS spectra of fluorocarbon films on SiO_2 surfaces: (a) C 1s and (b) F 1s, derived from n-$C_3F_7OCH_3$ (HFE-347mcc3), i-$C_3F_7OCH_3$ (HFE-347mmy), and $CF_3CF_2CF_2CH_2OH$ (PPC) plasmas. Reproduced with permission from [43], Copyright 2023, American Chemical Society.

compared to the C_4F_8/H_2 and $C_3H_3F_3$ plasmas, due to a higher F/C ratio of fluorocarbons on the SiO_2 surface. Furthermore, the SiO_2 ALE process utilizing C_4F_8, CHF_3, and $C_3F_7OCH_3$ plasmas has been examined to assess the effects of both hydrogen and oxygen additions [8]. The fluorocarbon film formed by the $C_3F_7OCH_3$ plasma exhibited the lowest F/C ratio compared to those produced by the C_4F_8 and CHF_3 plasmas.

The SiO_2 ALE process utilizing $C_4H_3F_7O$ isomer plasmas was explored to elucidate the impact of chemical structure [43]. The composition of the resulting fluorocarbon film is dictated by the chemical bonding structure of the $C_4H_3F_7O$ isomers, leading to variations in the F/C ratios as shown in Fig. 2. The F/C ratios of fluorocarbons generated by n-$C_3F_7OCH_3$ and i-$C_3F_7OCH_3$ plasmas are lower compared to those formed by $CF_3CF_2CF_2CH_2OH$ plasma, which can be attributed to the CH_3 radicals stemming from the presence of -OCH_3 in the molecule. Correspondingly, the etching rate of SiO_2 is highest in the $CF_3CF_2CF_2CH_2OH$ plasma and lowest in the i-$C_3F_7OCH_3$ plasma, mirroring the F/C ratio of the fluorocarbon films. It is observed that the addition of hydrogen to the plasma reduces the F/C ratio of fluorocarbon films and subsequently diminishes the etching rate of SiO_2.

2.2. Ion energy: Threshold energy and ALE window

In the anisotropic SiO_2 ALE process, the parameters defining the ALE window include the precursor used in the fluorination step, the type of gas employed in the removal step, and the ion energy, as summarized in Table II. The threshold physical sputtering energies identified are 50 eV for SiO_2 [24], 20 eV for Si_3N_4 [23], and 20 eV for Si [24]. Investigations into the physical sputtering of SiO_2, based on the Ar bias voltage, revealed sputtering at ion energies above 65 eV [27]. Surface modification allows for etching with lower ion energy and enables self-limiting removal of the modified layer [7]. In the ALE window region, selective removal of the modified layer from the bulk material

Table II. Ion energy windows in anisotropic ALE of SiO_2.

Precursor chemistries for fluorination	Removal	Ion energy in the removal step (Bias voltage)	Process temperature (°C)	Etching rate (Å/cycle)	Ref.
CHF_3 plasma	Ar plasma	0–50V	−40–20	9.0–11.0	[33]
C_4F_8 plasma beam	Ar ion beam	30–200V	RT	1.9	[27]
C_4F_6/Ar plasma	Ar plasma	10–100V	−10	14.2	[40]
C_4F_8 plasma CHF_3 plasma $C_3F_7OCH_3$ plasma	Ar plasma O_2 plasma	30–90V	20	5.8 4.1 2.1	[8]
n-$C_3F_7OCH_3$ plasma i-$C_3F_7OCH_3$ plasma $CF_3CF_2CF_2CH_2OH$ plasma	Ar plasma	10–80V	RT	2.1 1.8 5.2	[43]

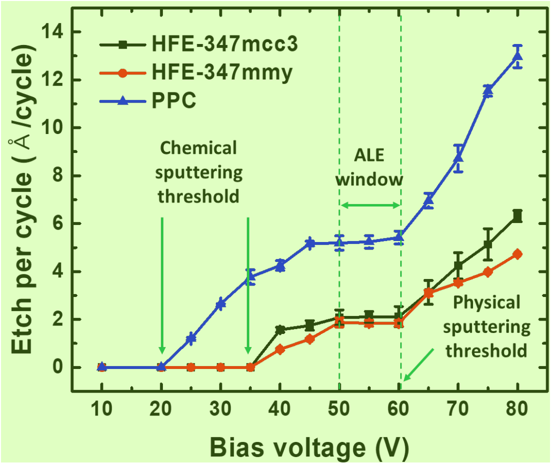

Figure 3. SiO_2 etching rates correlated with bias voltage in etching step, highlighting incomplete etching, ALE window, and physical sputtering regions based on ion energy. Reproduced with permission from [43], Copyright 2023, American Chemical Society.

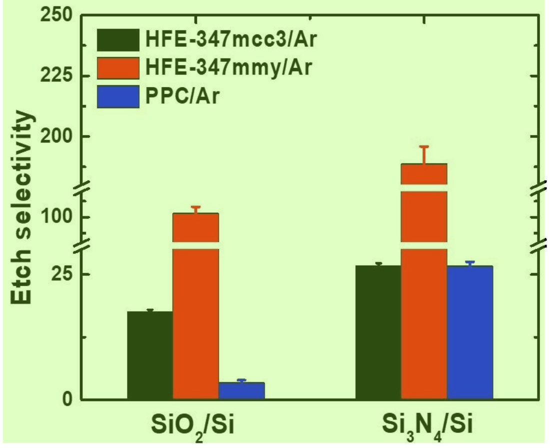

Figure 4. Precursor-dependent etching selectivity analysis for SiO_2/Si and Si_3N_4/Si in fluorination step. Reproduced with permission from [43], Copyright 2023, American Chemical Society.

occurs. Three distinct regions are discerned based on ion energy: the incomplete etching region, the ALE window region, and the physical sputtering region, as shown in Fig. 3 [43]. The removal of the modified layer is not complete at low ion energies, while sputtering of the underlying material occurs at higher ion energies.

The SiO_2 ALE window region has been reported with varying parameters. A previous study reported a window region of 50–60 V with 15 W source power of Ar plasma [8]. Another study identified an SiO_2 ALE window region with a bias voltage of approximately 15 eV at 1 kW of Ar source plasma power, highlighting a very low ion energy range for the ALE window due to the high source power [42]. These findings imply that interactions between chemical structure, ion energy, and process conditions determine the efficacy and characteristics of SiO_2 ALE processes.

2.3. Selectivity

The etching selectivity of SiO_2/Si and SiO_2/Si_3N_4 can be enhanced through various approaches, including the choice of precursors, control of fluorocarbon film thickness, ion energy management, etch step time adjustment, and selective deposition techniques, as outlined in Table III. The selectivity in anisotropic SiO_2 ALE is influenced by the thickness of the fluorocarbon film, which in turn is determined by the choice of precursor [23]. Improvements in SiO_2/Si and SiO_2/Si_3N_4 etch selectivity have been achieved using C_4F_8, C_4F_8/H_2, and $C_3H_3F_3$ plasmas by studying the impact of hydrogen addition [31]. The etching selectivity of SiO_2/Si and SiO_2/Si_3N_4 were notably improved using the $C_3H_3F_3$ precursor, attributed to the reduction of fluorine concentration in the fluorocarbon film due to hydrogen. Additionally, the enhancement of SiO_2/Si etch selectivity has been discussed using $C_4H_3F_7O$ isomer plasma, as shown in Fig. 4 [43]. A higher presence of Si-C bonds was observed in the fluorocarbon films generated by the i-$C_3F_7OCH_3$ plasma, compared to those from n-$C_3F_7OCH_3$ and $CF_3CF_2CF_2CH_2OH$ plasmas. These Si-C bonds act as inhibitors for Si etching, thereby decreasing the etching rate of Si and enhancing the SiO_2/Si etch selectivity. This evidence suggests that carefully selecting precursors can lead to high etch selectivity in anisotropic ALE processes. Moreover, selective functionalization of the SiN_x surface with benzaldehyde has been shown to improve the etch selectivity of SiO_2/SiN_x, as shown in Fig. 5 [38]. Benzaldehyde selectively deposits on SiN_x surfaces featuring -NH_x (x = 1, 2) groups, but not on SiO_2 surfaces with -OH groups. This selective deposition of benzaldehyde on SiN_x surfaces fosters the formation of a hydrofluorocarbon film, which serves as a barrier against the etching of SiN_x. The implication of these findings is that through strategic precursor selection and surface functionalization, the etching selectivity for different material combinations in anisotropic ALE processes can be effectively manipulated and optimized.

Table III. Studies of etch rate selectivity in anisotropic ALE of SiO_2.

Selectivity of material	Selectivity	Improving etch selectivity method	Precursor chemistries for fluorination	Removal	Ref.
SiO_2/Si_3N_4	0.2-15.0	Precursor selection, Fluorocarbon film thickness, Ion energy, Etching step time	C_4F_8/Ar plasma CHF_3/Ar plasma	Ar plasma	[24]
SiO_2/Si_3N_4 SiO_2/Si	>7.0 >10.0	Precursor selection	C_4F_8/Ar plasma C_4F_8/H_2/Ar plasma $C_3H_3F_3$/Ar plasma	Ar plasma	[31]
SiO_2/Si	2.6-17.5	Precursor selection	C_4F_8 plasma CHF_3 plasma $C_3F_7OCH_3$ plasma	Ar plasma O_2 plasma	[8]
SiO_2/Si	17.5 102.8 3.4	Precursor selection	n-$C_3F_7OCH_3$ plasma i-$C_3F_7OCH_3$ plasma $CF_3CF_2CF_2CH_2OH$ plasma	Ar plasma	[43]
SiO_2/Si_3N_4	2.1-4.5	Selective deposition (benzaldehyde)	C_4F_8/Ar plasma	Ar plasma	[38]

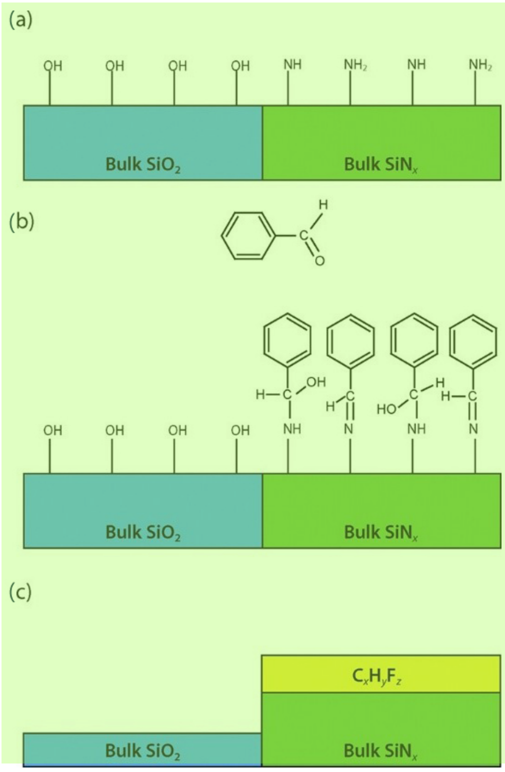

Figure 5. Selective ALE process of SiO_2 over SiN_x: (a) Initial functional groups on plasma-deposited surfaces; (b) selective functionalization of SiN_x surface with benzaldehyde; (c) aromatic aldehyde aiding graphitic hydrofluorocarbon film formation on SiN_x surface during ALE, inhibiting SiN_x etching. Reproduced with permission from [38], Copyright 2021, AIP Publishing.

2.4. Chamber wall effect

In the anisotropic SiO_2 ALE process, the etching rate is influenced by the fluorocarbon film left on chamber walls, as summarized in Table IV. Achieving a consistent etching rate is crucial in ALE and the fluorocarbon film on the chamber walls plays a significant role in the repeatability of the etching rate. Variations in the SiO_2 etching rate across multiple ALE cycles have been reported using *in situ* ellipsometry [26]. It was observed that a fluorocarbon film deposited on the chamber wall leads to an increase in the etching rate as the cycle repetition increased. The impacts of the quartz window temperature and a fluorocarbon film on the chamber wall on the etching rate was reported as shown in Fig. 6 [22]. Research focusing on mitigating the chamber wall effect has been conducted using O_2 plasma to maintain a constant etching rate [29]. The application of O_2 plasma effectively prevented the buildup of a fluorocarbon film on the chamber walls, thereby preserving the initial state of the chamber.

Table IV. Studies on impact of chamber wall conditions in anisotropic ALE of SiO_2.

Precursor chemistries for fluorination	Removal	Method of removal chamber wall effect	Etching rate (Å/cycle)	Ref.
C_4F_8/Ar plasma	O_2 plasma	O_2 plasma	5.6–11.4	[29]
C_4F_8/Ar plasma	Ar plasma	Chamber cleaning with O_2 plasma, Chamber wall heating	3.0	[22]

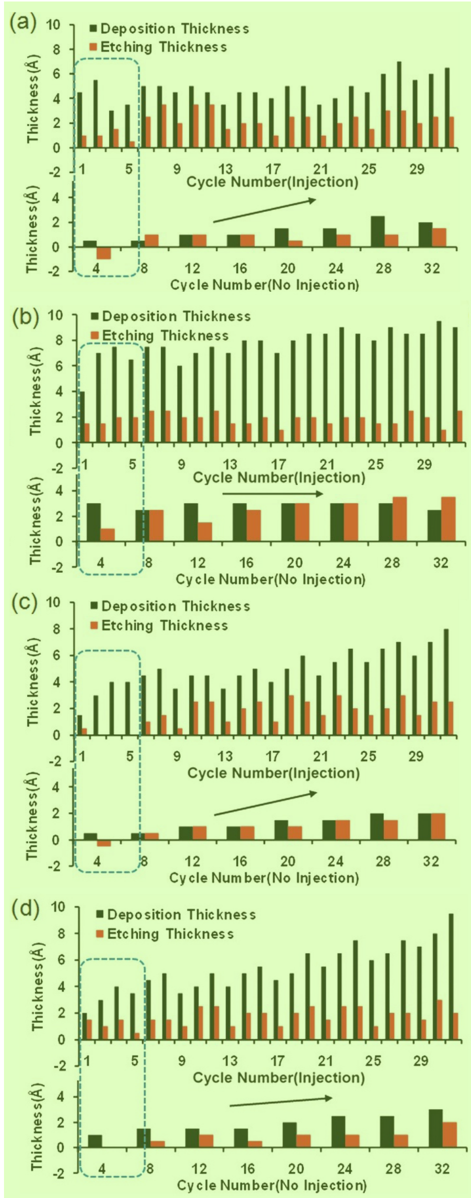

Figure 6. Analysis of deposition and etching thickness under four conditions: (a) Cold/clean, (b) cold/with film, (c) hot/clean, and (d) hot/with film. Reproduced with permission from [22], Copyright 2016, AIP Publishing.

3. Isotropic SiO_2 ALE

3.1. Thermal isotropic ALE

In isotropic SiO_2 ALE processes, the SiO_2 surface undergoes modification through either thermal reactions or plasma-assisted methods. The etching mechanisms for thermal isotropic SiO_2 ALE are summarized in Table V. Here, the SiO_2 surfaces are modified by conversion to Al_2O_3 or by formation of ammonium salt, and then the modified layers are removed through various chemical reactions including sublimation, fluorination, and ligand exchange. These etching mechanisms are categorized into two groups: conversion of SiO_2 into Al_2O_3 or ammonium fluorosilicate (AFS) during the modification step. An example of the isotropic SiO_2 ALE process using triethylaluminium (TMA) and hydrogen fluoride (HF) precursors was reported [25]. In this process, SiO_2 is converted to Al_2O_3 utilizing the TMA precursor, as indicated in Eq. (1), which represents the reaction for modification

Table V. Studies on thermal isotropic ALE of SiO_2.

Etching mechanism	1st step	2nd step	3rd step	Etching rate (Å/cycle)	Ref.
Conversion: $SiO_2 \rightarrow Al_2O_3$ Fluorination: $Al_2O_3 \rightarrow AlF_3$ Ligand exchange: $AlF_3 \rightarrow AlF(CH_3)_2$	TMA (300 °C)	HF (300 °C)	TMA (300 °C)	0.07–0.27	[25]
	TMA (350 °C)	HF (350 °C)	TMA (300 °C)	0.35–1.50	[32]
Conversion: $SiO_2 \rightarrow (NH_4)_2SiF_6$ Heating: $(NH_4)_2SiF_6 \rightarrow SiF_4 + 2NH_3 + 2HF$	HF (20 °C)	NH_3 (20 °C)	Heating (140 °C)	9	[42]

Table VI. Studies on plasma-assisted isotropic ALE of SiO_2.

Etching mechanism	1st step	2nd step	3rd step	Etching rate (Å/cycle)	Ref.
Conversion: $SiO_2 \rightarrow (NH_4)_2SiF_6$ Heating: $(NH_4)_2SiF_6 \rightarrow SiF_4 + 2NH_3 + 2HF$	CF_4/NH_3 plasma NF_3/NH_3 plasma (20 °C)	Heating (160 °C)	–	27–70	[35]
	NF_3/H_2 plasma (20 °C)	NH_3 (20 °C)	Heating (150 °C)	75	[39]

step. Subsequently, Al_2O_3 undergoes fluorination to AlF_3 by HF, as shown in Eq. (2). Finally, AlF_3 is removed via a ligand exchange reaction forming volatile $AlF(CH_3)_2$ with TMA, as shown in Eq. (3), which represents the reaction for removal step. These reactions occur spontaneously at temperatures as high as 300 °C without plasma assistance.

$$3SiO_2 + 4Al(CH_3)_3 \rightarrow 2Al_2O_3 + 3Si(CH_3)_4. \quad (1)$$
$$Al_2O_3 + 6HF \rightarrow 2AlF_3 + 3H_2O. \quad (2)$$
$$AlF_3 + 2Al(CH_3)_3 \rightarrow 3AlF(CH_3)_2. \quad (3)$$

Isotropic SiO_2 ALE process using HF and NH_3 gas has been also reported [42]. The AFS layer is formed by the reaction of NH_3 molecules with the adsorbed HF on SiO_2 surface, as shown in Eq. (4). In the removal step, the AFS layer decomposes at temperatures above 140 °C, forming volatile reaction products such as NH_3, HF, and SiF_4, as shown in Eq. (5).

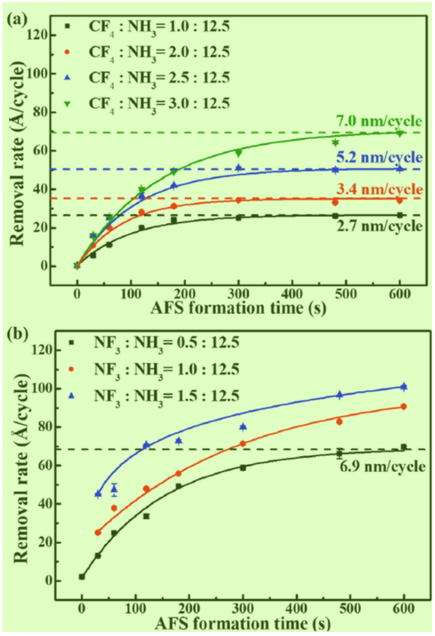

Figure 7. AFS formation time-dependent removal rates with (a) CF_4/NH_3 plasma and (b) NF_3/NH_3 plasma. Reproduced with permission from [35], Copyright 2020, AIP Publishing.

$$SiO_2 + 6HF + 2NH_3 \rightarrow (NH_4)_2SiF_6 + 2H_2O. \quad (4)$$
$$(NH_4)_2SiF_6 \rightarrow 2NH_3 + 2HF + SiF_4. \quad (5)$$

3.2. Plasma-assisted isotropic ALE

The etching mechanisms of plasma-assisted isotropic SiO_2 ALE are comprehensively summarized in Table VI. A specific isotropic SiO_2 ALE process employing CF_4/NH_3 or NF_3/NH_3 plasma has been examined. In this process, SiO_2 is converted to AFS in CF_4/NH_3 or NF_3/NH_3 plasma as shown in Eqs. (6) and (7). Subsequently, the AFS layer decomposes at temperatures above 160 °C, forming volatile substances such as NH_3, HF, and SiF_4, as shown in Eq. (8). The self-limiting property of AFS formation was confirmed to be dependent on the plasma duration, with the etching rate escalating from 2.7 to 7.0 nm/cycle based on the gas ratio, as shown in Fig. 7. The etching rate observed in the NF_3/NH_3 plasma was approximately threefold higher than that in the CF_4/NH_3 plasma. This discrepancy is attributed to the different bonding energies of F (~506 kJ/mol) and N-F (~239 kJ/mol). NF_3 is more prone to dissociation than CF_4 under similar conditions, leading to the generation of a greater number of fluorine atoms in the NF_3/NH_3 plasma compared to the CF_4/NH_3 plasma. This results in more extensive AFS formation and consequently, a higher etching rate. This step is crucial as it ensures the complete removal of the modified layer, enabling the process to proceed to the next cycle effectively. The distinction in etching rates between the two plasma types underscores the importance of gas selection and plasma conditions in optimizing the isotropic ALE process for SiO_2.

$$NH_3 + HF \rightarrow NH_4F. \quad (6)$$
$$SiO_2 + 4HF + 2NH_4F \rightarrow (NH_4)_2SiF_6 + 2H_2O. \quad (7)$$
$$(NH_4)_2SiF_6 \rightarrow 2NH_3 + 2HF + SiF_4. \quad (8)$$

4. Conclusions

In this review, we categorized recent research on the SiO_2 ALE process into anisotropic and isotropic processes. For anisotropic SiO_2 ALE processes, the effects of the precursor, ion energy, selectivity, and chamber wall conditions were examined. The choice of precursor influenced the F/C ratio in the film deposited on the SiO_2 surface, which subsequently affected the etching rate and selectivity. In the anisotropic ALE process, changes in ion energy affected the etching rate, and a consistent etching rate was observed within the defined ALE window. For isotropic SiO_2 ALE, we elucidated two types of mechanisms are summarized. SiO_2 surface into a fluorinated layer or ammonium salt, followed by removal through various chemical reactions, including sublimation, fluorination, and ligand exchange. Given the increasing complexity and three-dimensional nature of semiconductor device integration, the necessity for both anisotropic and isotropic ALE processes is evident. The selection of specific SiO_2 ALE mechanisms should be tailored to the device architecture. Future research on the nuances of the SiO_2 ALE mechanism is essential to further enhance the precision and efficiency of these processes in semiconductor fabrication.

Acknowledgments

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Conflict of interest

The authors declare no conflicts of interest.

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